Income Source and Income Change Variables

Preliminary Analysis

Needs Assessment & Targeting Unit at WFP APP HQ

May 20, 2024

Background

- The income source and income change variables have been used specifically since COVID-19 in WFP as a proxy indicator for economic vulnerability in current coping capacity.

- This presentation explains how the module has been implemented and

- Distribution of the income using data collected in Myanmar, Ecuador and Central African Republic (CAR).

Variable	Questions in Income Module	Choice List
HHHOccupation	What is the occupation of the household head?	Open Ended
HHInc_PCT	How many household members are contributing to the household income?	Number
HHIncNb	How many income sources does your household have?	Number
HHIncFirst	Main source of income for the household.](Hint: This is the most stable income source for the household)Relevance: HHIncNb > 0	1 Wage Labor – Professional 2 Wage Labor – Skilled 3 Wage Labor - Unskilled/Casual/Agriculture 4 Wage Labor - Unskilled/Casual/non-agriculture 5 Pension 6 Remittances 7 Aid/gifts 8 Borrowing money/Living off debt 9 High risk activity (e.g. begging, scavenging) 10 Saving/selling assets 11 Petty trade/selling on streets 12 Small trade (own business) 13 Medium/large trade (own business) 14 Small Agriculture production including livestock (own land/livestock) 15 Medium/large agriculture production including livestock (own land/livestock) 999 Other (specify)
HHIncFirst_Oth	SpecifyRelevance: HHIncFirst_SRi = '999'	
HHIncChg	Was there a change in income over the last year? (Please do not consider currency devaluation.)	0 No change 1 Increased 2 Reduced by less than 25% 3 Reduced by more than 25% and less than 50% 4 Reduced by more than 50%

Categorization of Income Sources & Change

Income Sources:

- **Regular employment/income**
  - Wage Labor – Professional/skilled
  - Pension
  - Small/Medium/large trade (own business)
  - Small/Medium/Large agriculture production including livestock (own or rented land/owned livestock)
- **Irregular employment/income**
  - Wage Labor - Unskilled/casual/agriculture/unskilled/casual/non-agriculture
  - Remittances
  - Petty trade/selling on streets
- **No income**
  - Aid/gifts
  - Borrowing money/living off debt
  - High risk activity (e.g., begging, scavenging)
  - Saving/selling assets
  - No income source noted in *HHIncNb*

Income change is categorized as increased, reduced and no change.

Aggregated Income Indicator

The combination of main income sources and changes over the past 12 months further aggregates to an **overall 4-point income indicator**

Indicator	Food Secure	Marginally Food Secure		Moderately Insecure	Severely Insecure
Main income source	Regular employment or income	Regular employment or income	Irregular employment or income	Irregular employment or income	No income
Income Change	Increased income and no change	Reduced income	Increased income and no change	Reduced income	

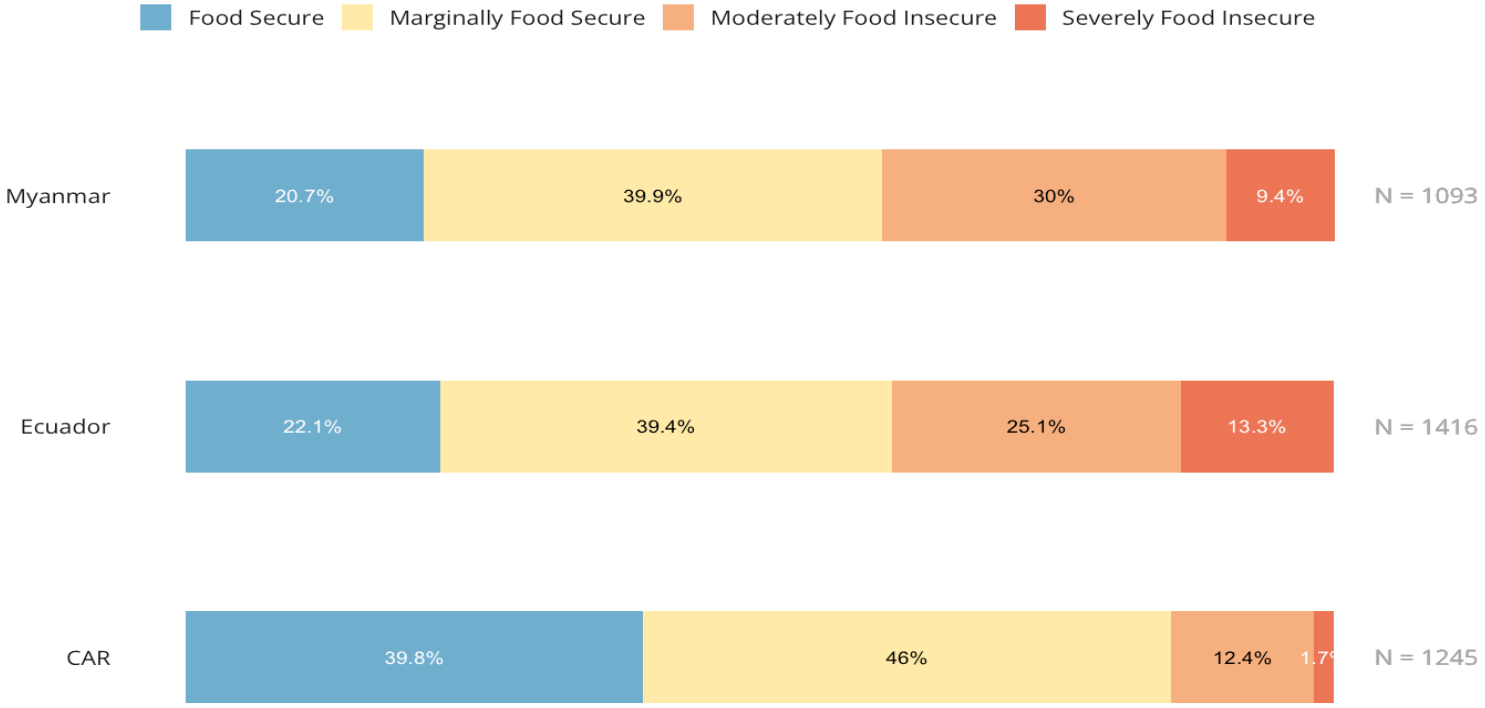
Income Scripts

Economic vulnerability: Income

```
ren HHIncFirst_Oth HHIncFirst_oth
replace HHIncFirst_oth = strproper(strtrim(HHIncFirst_oth))
tab HHIncFirst_SRi
tab HHIncFirst_SRi
gen       HHIncFirst_Accept = inlist(HHIncFirst_SRi, 1, 2, 5, 13, 14, 15, 16)
replace HHIncFirst_Accept = 1 if HHIncFirst_oth == "Pasteur"
gen             HHIncFirst_Border = inlist(HHIncFirst_SRi, 3, 4, 6, 12)
replace HHIncFirst_Border = 1 if HHIncFirst_oth == "Coturier" | HHIncFirst_oth == "Cultuvateure" |
HHIncFirst_oth == "Moto Taxi"
gen       HHIncFirst_Poor         = inlist(HHIncFirst_SRi, 7, 8, 9, 10, 11)
gen CARI_Inc = .
replace CARI_Inc = 1 if HHIncFirst_Accept == 1 & inlist(HHIncChg, 0, 1, 5)
replace CARI_Inc = 2 if HHIncFirst_Accept == 1 & inlist(HHIncChg, 2, 3, 4)
replace CARI_Inc = 2 if HHIncFirst_Border == 1 & inlist(HHIncChg, 0, 1, 5)
replace CARI_Inc = 3 if HHIncFirst_Border == 1 & inlist(HHIncChg, 2, 3, 4)
replace CARI_Inc = 4 if HHIncNb == 0 | HHIncFirst_Poor == 1

tab CARI_Inc, gen(CARI_Inc_)
```

Household Economic Vulnerability Income Indicator by Country



Source: rCARI Validation Study, data collected 2023

Recategorization of Smallholder Farmers

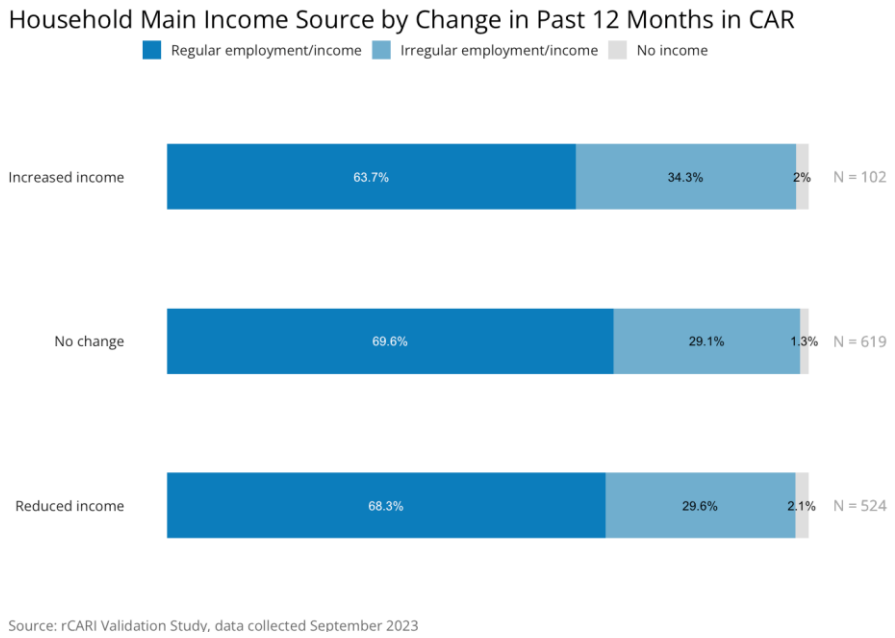
- Current income change indicators shows downward bias because of possible misclassification of low-income earners
  - **Small-scale farmers and small enterprise owners** in the 4-point scale into food secure.
- Re-categorizing the small-scale farmers and small enterprise owners to Irregular Income Group are presented as below:

	Country		
Income Variables	**Myanmar**, N = 1,093[1]	**Ecuador**, N = 1,416[1]	**CAR**, N = 1,245[1]
HHIncome_3pt_Re			
Regular employment/income	203(19%)	369(26%)	478(38%)
Irregular employment/income	787(72%)	858(61%)	672(54%)
No income	103(9.4%)	189(13%)	95(7.6%)
HHIncChg_3pt			
Increased income	141(13%)	50(3.5%)	102(8.2%)
No change	453(41%)	712(50%)	619(50%)
Reduced income	499(46%)	654(46%)	524(42%)

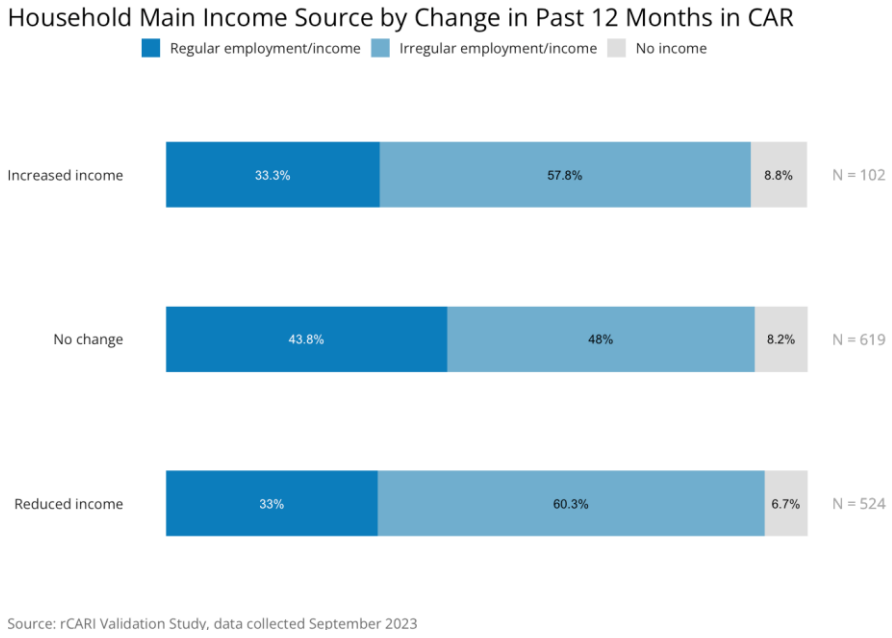
[1] n(%)

Before vs. After Recategorization - CAR

Before



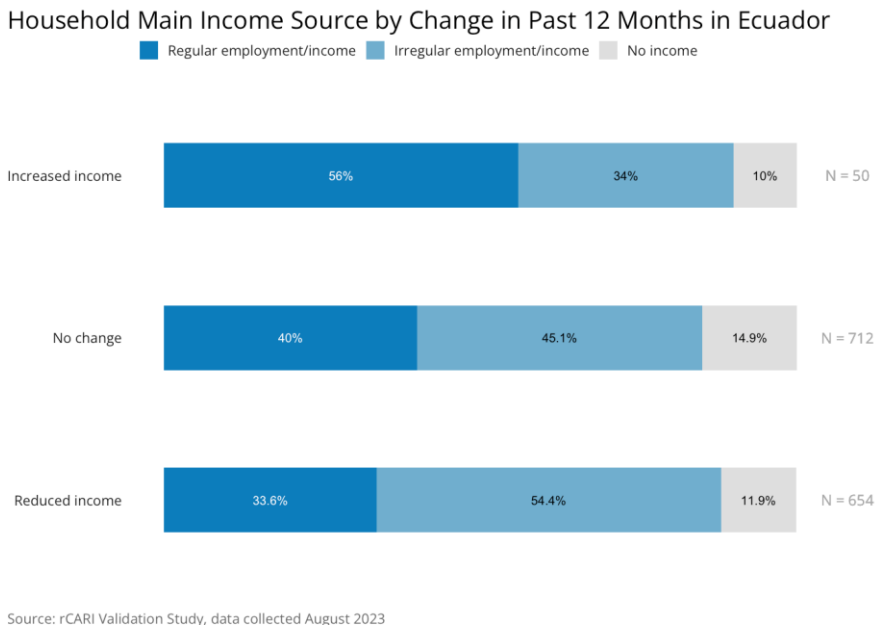
After



Note: The other specify answers in income source variable (HHIncFirst_SRi) are fully cleaned in the aggregated 3pt category.

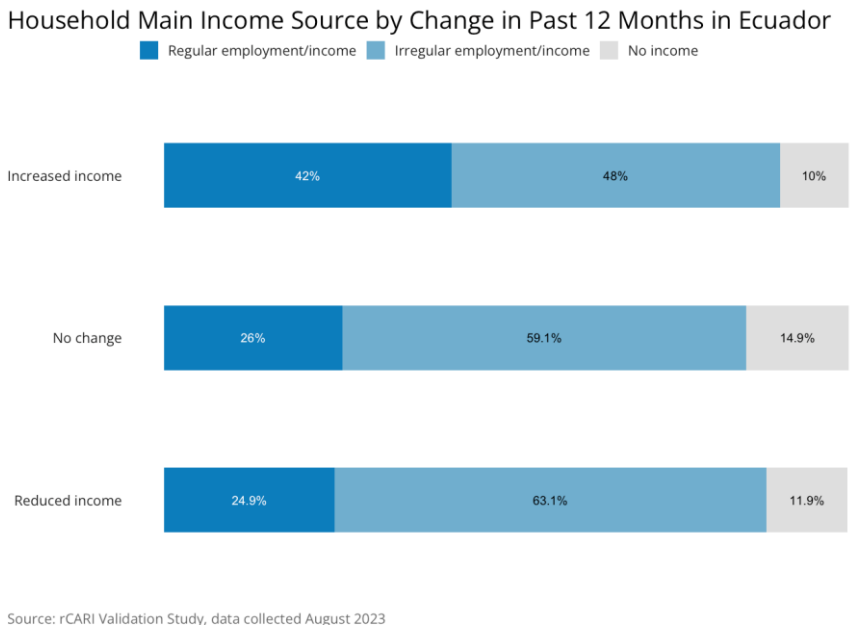
Before vs. After Recategorization - Ecuador

Before



Source: rCARI Validation Study, data collected August 2023

After

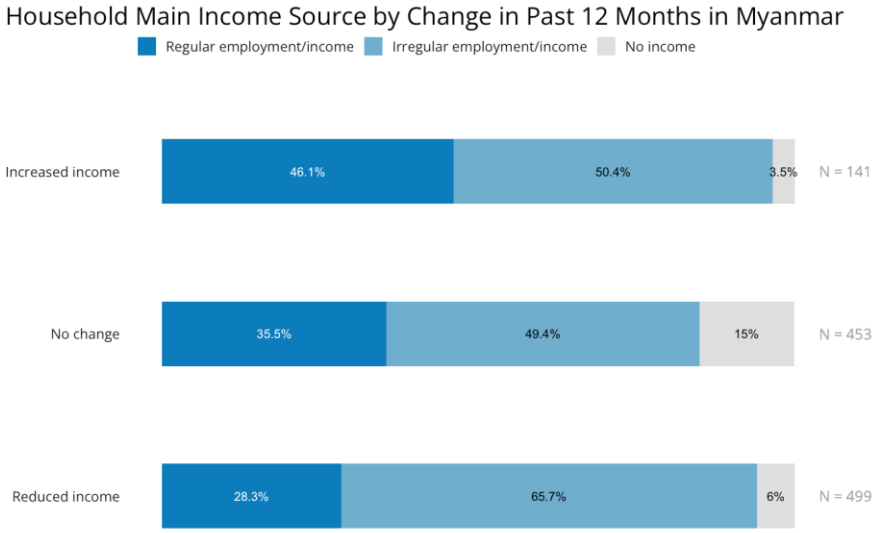


Source: rCARI Validation Study, data collected August 2023

Note: The other specify answers in income source variable (HHIncFirst_SRi) are fully cleaned in the aggregated 3pt category.

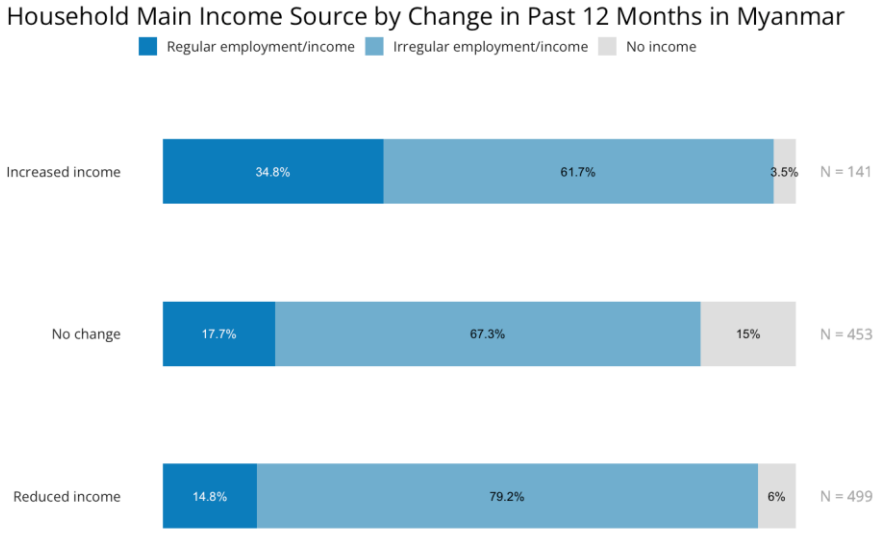
Before vs. After Recategorization - Myanmar

Before



Source: rCARI Validation Study, data collected June 2023

After



Source: rCARI Validation Study, data collected June 2023

Note: The other specify answers in income source variable (HHIncFirst_SRi) are fully cleaned in the aggregated 3pt category.

Next Steps

- Expand the testing to other available dataset to enable larger evidence base for testing the indicator.
- Possible reclassification of the income types based on the standardized income categorization of Lower-, Lower-middle-, Middle-, and High-income groups.
- Sensitivity analysis of the individual income sources [HHIncFirst] based on the results obtained from the correlation and sensitivity analysis.

Thank You!